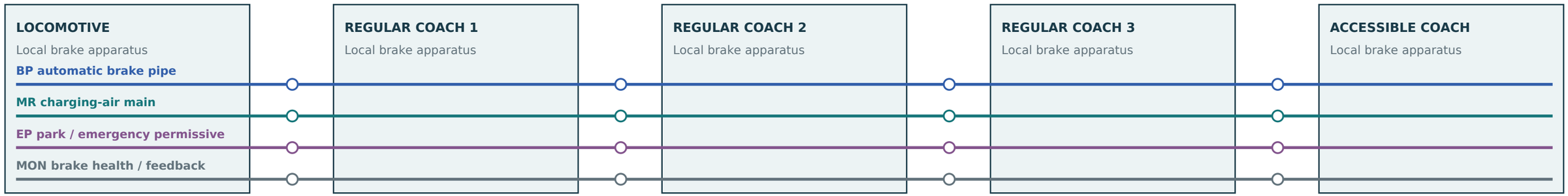
PULSIPHER FARMS / CHRISTMAS RAILWAY

# AUTOMATIC AIR BRAKES / BETWEEN-CAR INTERFACES

Candidate A functional architecture. Each function needs a documented valve realization; lines below are NOT installation plumbing or a safety validation.

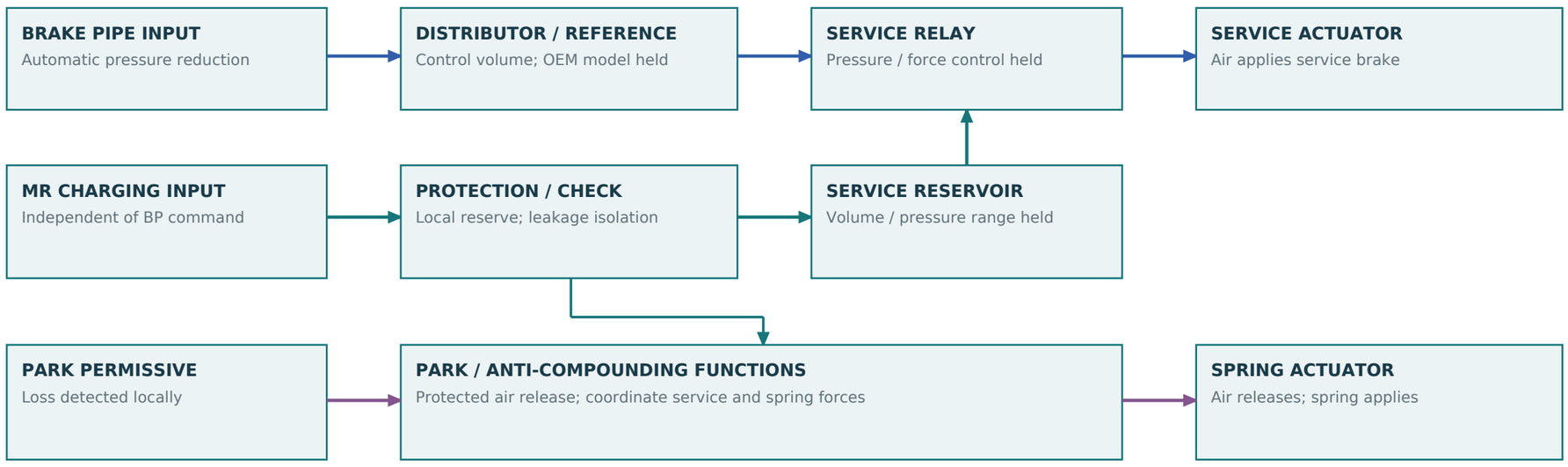
## K1.1 FULL CONSIST / FOUR DISTINCT INTERFACE FUNCTIONS / NOT TO SCALE




Four intervehicle joints. BP pressure reduction commands service braking; protected local air supplies the actuators.

EP + MON are electrical functions, not air hoses. Connector sharing, circuits, voltage, timing and fault logic remain HELD.

## K1.2 ONE CAR / REQUIRED FUNCTIONS / ARROWS ARE FUNCTIONAL RELATIONSHIPS




Local emergency functions: vent BP, inhibit traction and arbitrate spring/service forces. BP rupture with MR intact must command stored-air service braking; it does not by itself prove spring application. Actual response requires testing.

## K1.3 EACH PHYSICAL JOINT

BP: rail-compatible brake-pipe hose / coupling and isolation arrangement; continuity check at the rear.

MR: separately identified supply-air hose / coupling with protection against a failed car or separated train.

Electrical: EP continuity plus MON feedback. Shared connector requires segregation analysis. Base BP braking must not depend on a data network.

HELD: hose ID, length, pressure rating, bend radius, port fittings, valves, connector pinout, clearance and retention.

# COMBINATION AIR CHAMBER / DOCUMENTED ENVELOPE

MGM TR3030, P/N 3430051 sourcing candidate. Nominal OEM dimensions; tolerances, bracket design and rail compatibility are not established.

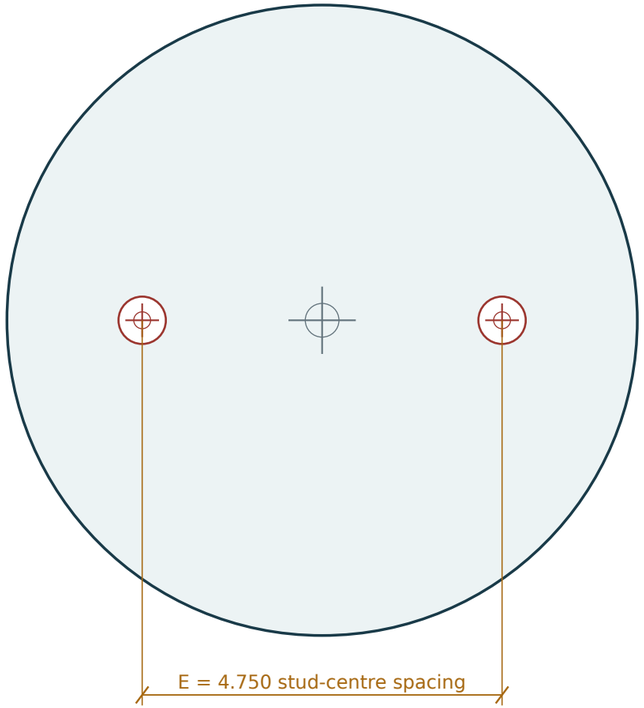
## K2.1 NOMINAL MAXIMUM CASING ENVELOPE / SIDE VIEW / INCHES




Red reference: actuator mounting face. Housing box excludes rod, studs, air fittings, release tool and required service / inspection clearance.

OEM manual EB24-001 Rev02 p11: pushrod within +/-3 degrees through stroke; drain within 30 degrees of bottom. Actual bracket, ports, clevis and service-access geometry remain held.

## K2.2 MOUNT-FACE ENVELOPE / STUD AXES / INCHES




**Studs**

5/8-11 UNC
Length 1.540 in

**Pushrod**

5/8 in diameter
5/8-18 UNF thread
8 in stock; no yoke

**Air ports**

3/8-18 NPTF
Position / clocking held

## K2.3 OEM FACTS AND USE LIMITS

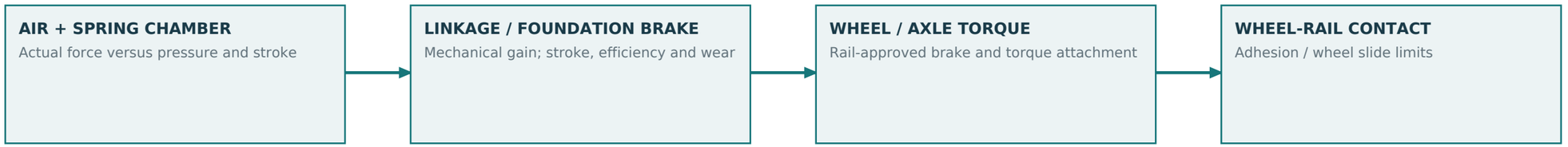
| Item | Documented fact / limit |
|---|---|
| Rated stroke | 2.500 in; NOT allowable operating stroke |
| Spring case OD / mass | D = 8.210 in / 19.4 lb combination |
| B = 6.710 in | OEM figure datum retained in source; not modeled |
| Long-stroke alternative | 3230551 candidate; exact envelope not substituted |

Source: MGM Form 5019, p2 + current 3430051 product record; retrieved 2026-10-19 UTC.

# BRAKE DEMAND / FOUNDATION BRAKE INTERFACE

Calculations are conditional screens, not demonstrated stopping distances. Mechanical brake selection, friction, fade, adhesion and load balance remain open.

## K3.1 FORCE PATH / ONE BRAKED AXLE / NOT TO SCALE




Train slows only if usable rail contact force remains available.

Reaction path: actuator / brake mount -> bogie -> axle / suspension restraint -> body connection. Every attachment needs loads.

## K3.2 CANONICAL ANALYTICAL CASE / ALL VEHICLES BRAKED

| Input / output | Value | Meaning / limitation |
|---|---|---|
| Whole train mass | 65,100 kg upper study | 3 x 13,200 + 11,000 + 14,500 kg |
| Grade / assumed deceleration | 0% target / 0.500 m/s2 | Achieved survey and rail condition unverified |
| Application delay | 1.000 s assumption | Instant full force after delay; no ramp model |
| Required net braking force | 32.550 kN train | F = m a; no assistance from rolling drag |
| 5 mph stop / 7.5 mph stop | 7.231 m / 14.594 m | d = v delay + v2/(2 a), conditional |
| Regular / accessible / loco demand | 6.600 / 5.500 / 7.250 kN | Mass-proportional screen; not brake allocation |

## K3.3 TORQUE PROBE / NOT A WHEEL OR BOGIE SELECTION

Using a 0.300 m rolling radius and FOUR braked axles per coach solely for a numerical probe: translation-only torque: regular 495.0 N m/axle; accessible 412.5 N m/axle. T = F_coach r / n_axles; rotating inertia is additional.

The actual wheel radius, number of brakes, effective disc / tread radius, friction, linkage, chamber stroke and normal axle loads must be supplied together.

## K3.4 REQUIRED COMPONENT RETURN / BEFORE FABRICATION

Document the complete rail foundation brake: torque versus input force and stroke; hot / cold friction and wear limits; park holding; bracket bolt pattern; fatigue loads; anti-compounding; inspectable adjustment; maximum stroke; wheel-slide protection or proven alternative.

No adapter plate thickness, lever length, pin, weld, clevis or rotor is assigned by this sheet. Those dimensions depend on the verified full load path.

# BRAKE BENCH / FIRST ARTICLE / FULL CONSIST TESTS

Test development only. Qualified engineer must approve the setup, loads, sequence, hazards, instrumentation and acceptance limits before physical testing.

## K4.1 SEQUENTIAL RELEASE POINTS / EACH RESULT CONTROLS THE NEXT STEP

**1 DOCUMENT REVIEW**
OEM matched components; force curves; interface loads; failure analysis

→

**2 GUARDED BENCH**
Air circuit + loaded actuator; normal operation and controlled faults

→

**3 UNLOADED VEHICLE**
Static fit, stroke, routing and leak checks; then controlled low-speed testing

→

**4 INSTRUMENTED CONSIST**
Ballasted loading; defined rail conditions; all vehicles and end-of-train

## K4.2 TEST MATRIX / RECORD ACTUAL RESPONSE, DO NOT INFER IT FROM A SCHEMATIC

| Test | Controlled condition | Required evidence | Acceptance basis - HELD |
|---|---|---|---|
| B01 | Service pressure / force / stroke | Calibrated BP, MR, reserve, chamber and force traces | OEM curves + accepted force / stroke window |
| B02 | BP rupture while MR stays charged | Rear BP drop; each local service response; no bypass | Accepted deceleration, delay and brake balance |
| B03 | MR loss / leakage / recharge | Reserve loss; repeat application; recovery; protection | Reserve / pressure / recharge limits from design |
| B04 | Permissive open / power loss | Local vent; traction inhibit; spring timing; compounding | Defined, verified response for every fault case |
| B05 | Park hold / release / isolated car | Force at worn condition; leak-down; release evidence | Accepted disturbance / hold load; no drag |
| V01 | Bogie articulation / suspension | Hose minimum bend and rub; stroke, clearances, retention | OEM limits at worst movement / wear combination |
| V02 | Full-consist stop and wheel slide | Mass, speed, pressure, time, distance, axle behavior | Approved speed / load / rail-condition envelope |
| V03 | Repeated stops / thermal recovery | Brake and bearing temperature; fade; pressure recovery | OEM thermal / wear / fade / recovery criteria |

## K4.3 TEST RECORD / ENGINEER TO FILL BEFORE TESTING

Article / consist: _____ Revision: _____ Engineer / test lead: _____ Date: _____

Calibration IDs: _____ Instrument locations: _____ Accepted limit sheet: _____ Test area: _____

Actual mass / load map: _____ Rail condition: _____ Wind / weather: _____ Deviation record: _____

Do not disassemble a spring chamber. Guard stored energy and use the actual OEM caging / handling procedure under a released test plan.